Week 7

Module 4 - A | Latent Variable Models

Expected Learning Outcomes

---------- Week 7 ----------

1: Concept of Clustering

2: Hard Clustering – K-means

3: Concept of Latent Variables

4: Expectation-Maximization (EM)

5: Soft Clustering – Mixture Models

---------- Week 8 ----------

6: Applications – EM for Gaussian Mixture Models (GMM)

7: Applications – EM for Document Clustering

Part 1: Concept of Clustering

Concept of Clustering

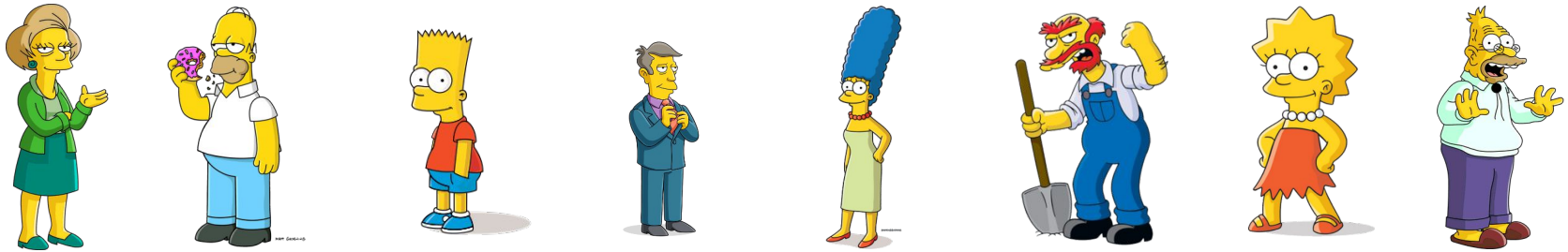
- Is a method of unsupervised learning
- Find a sensible structure from unlabelled data
- A clustering algorithm
  - Groups data into their natural categories
  - Without knowledge of their actual groups
  - Based on the similarities between them
  - Revealing the structure of the data

  >High intra-cluster similarity

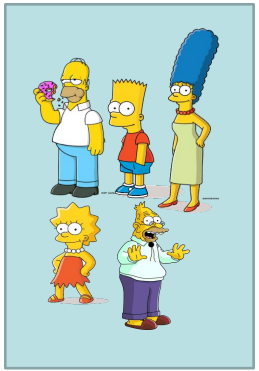
  >Low inter-cluster similarity

Concept of Clustering

- What is a natural grouping among these objects?



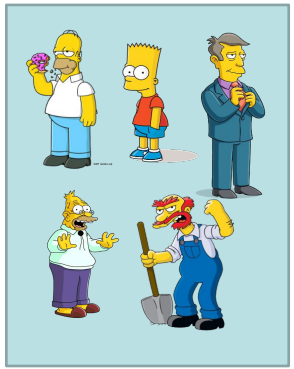
Clustering is subjective



Simpsons Family

School Employees

Females



Males

Clustering Method

1: K-Means clustering (hard clustering)

2: Mixture models (soft clustering)

3: Spectral clustering (e.g., Normalized cuts, Graph Laplacian…)

4: Biclustering

And so on

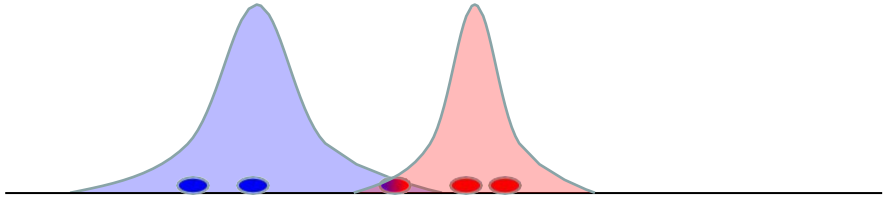
Bishop - Pattern Recognition and Machine Learning, 2006

Kevin Murphy – Probabilistic Machine Learning: An Introduction, 2023 (latest)

Soft vs. Hard Clustering

- Hard Clustering
  - Each data point belongs to only one cluster

- Soft Clustering
  - Data points could belong to one or more clusters
  - Probability of belonging to each cluster is needed to be estimated
  - Account for the uncertainty and ambiguity in data points' cluster memberships.



See Week 7 practice on FLUX for examples.

Part 2: K-Means for Hard Clustering

Hard Clustering: The K-Means Algorithm

- The simplest centre-based algorithm to solve hard clustering problems is K-Means

- N unlabelled data points $\{x_n\}$ are given
- Goal: Partition the data points into K distinct groups (clusters)
  - Similar points are grouped together
  - Similarity is based on a distance measure $d\,(.)$. Distance metrics can include Euclidean distance, Cosine distance, Wasserstein distance in optimal transport, Manhattan distance, Hamming distance, etc.

Hard Clustering: The K-Means Algorithm

- Is an iterative algorithm
- Starts with an initial random guess of K cluster centres

$$\left(\mu_1^{(0)}, \mu_2^{(0)}, \ldots, \mu_K^{(0)}\right)$$

- Iterate the following two steps until a stopping criterion is met:
  - Update assignment of data points to clusters
    - > Calculate the distance of each data point to all cluster centres
    - > Assign the data point to the cluster with the minimum distance
  - Update centers of the clusters
    - > For each cluster, calculate the new centre as the average of all data points assigned to it
    - > $$\mu_K^{(\tau+1)} = \frac{\sum_n r_{nk} x_n}{\sum_n r_{nk}}$$

$$r_{nk} = \begin{cases} 1 & \text{if } x_n \text{ is assigned to cluster } k \\ 0 & \text{Otherwise} \end{cases}$$

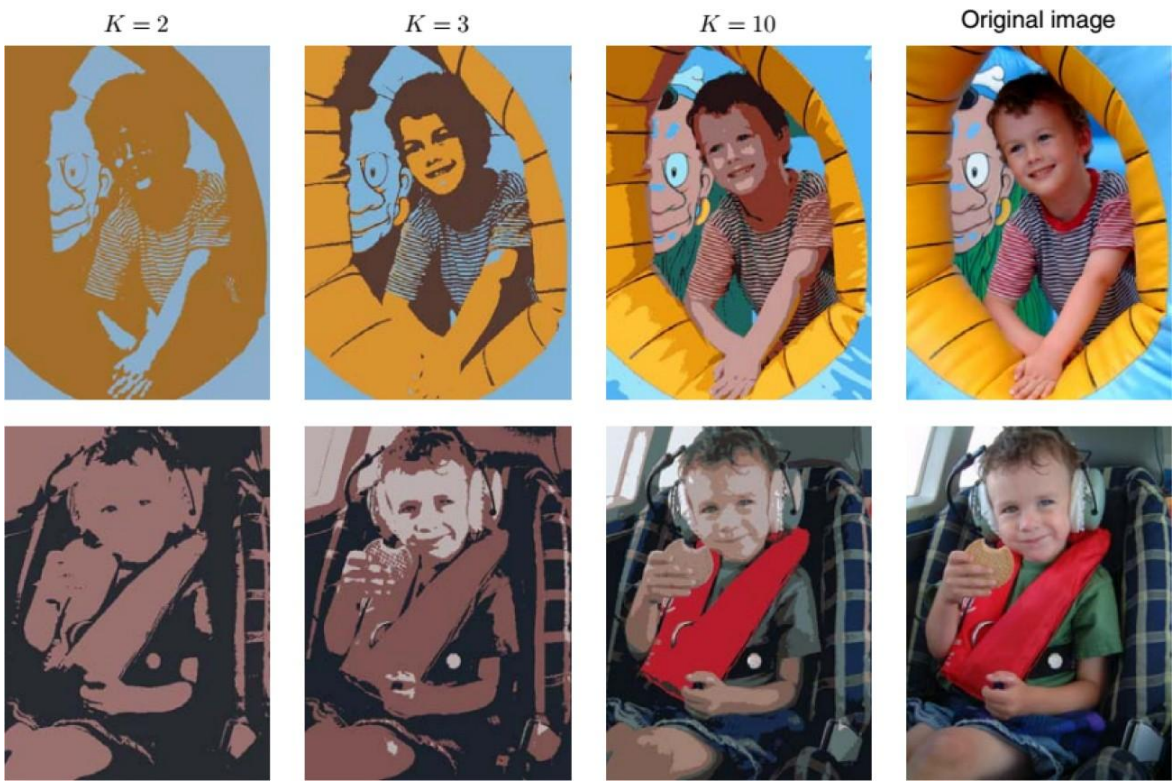
K-Means visualization

- A good visual simulation is available at http://tech.nitoyon.com/en/blog/2013/11/07/k-means/

K-Means Remarks

- K-Means is sensitive to initial values
  - which means the different execution of K-means with different initial cluster centers may result in different solutions
- K-Means is a non-probabilistic algorithm
  - which only supports *hard-assignment*
  - *a data point can only be assigned to one and only one of the clusters*

Applications of K-Means Clustering



- Data points: pixels colors
- Cluster: similar pixel colors
- Replace the colors in a cluster with the centroid
- Store the centroid only: reduced resolution and storage space

Part 3: Concept of Latent Variables

Big Picture for Soft Clustering

- In soft clustering models such as Gaussian Mixture Models (GMM), the latent variables are the cluster assignments. For each data point, the latent variable represents the probability distribution over clusters.

- These latent variables are used in probabilistic models to represent the relationship between observed data points and the underlying clusters. By estimating these latent variables, the model can determine the probabilities that each data point belongs to each cluster.

- In soft clustering methods, such as GMM, the estimation of latent variables often involves using algorithms like Expectation-Maximization (EM). The EM algorithm iteratively estimates the latent variables (E-step) and updates the model parameters (M-step) to maximize the likelihood of the observed data given the model.

Concept of Latent (Hidden) Variables

Latent variables are variables that are not directly observed but can be inferred based on other observed variables.

Examples:

In a recommendation system, a user's preferences for certain types of products may be latent variables (may because of missing or be confidential). But they can inferred based on their past behavior and other factors such as demographic information.

Clustering: In clustering algorithms like k-means, the cluster assignments of data points are latent variables. The algorithm tries to find the best assignment of points to clusters based on their similarities.

Recall Maximum Likelihood Estimation (MLE)

Definition: Estimate the parameters that assign the highest probability to the training data

Formulation:

A point estimate; ignore any notion of uncertainty in parameter estimate; without prior

$$\hat{\boldsymbol{\theta}}_{\text{mle}} \triangleq \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\mathcal{D} \mid \boldsymbol{\theta})$$

iid assumption (training samples are independent and identically distributed). The log likelihood:

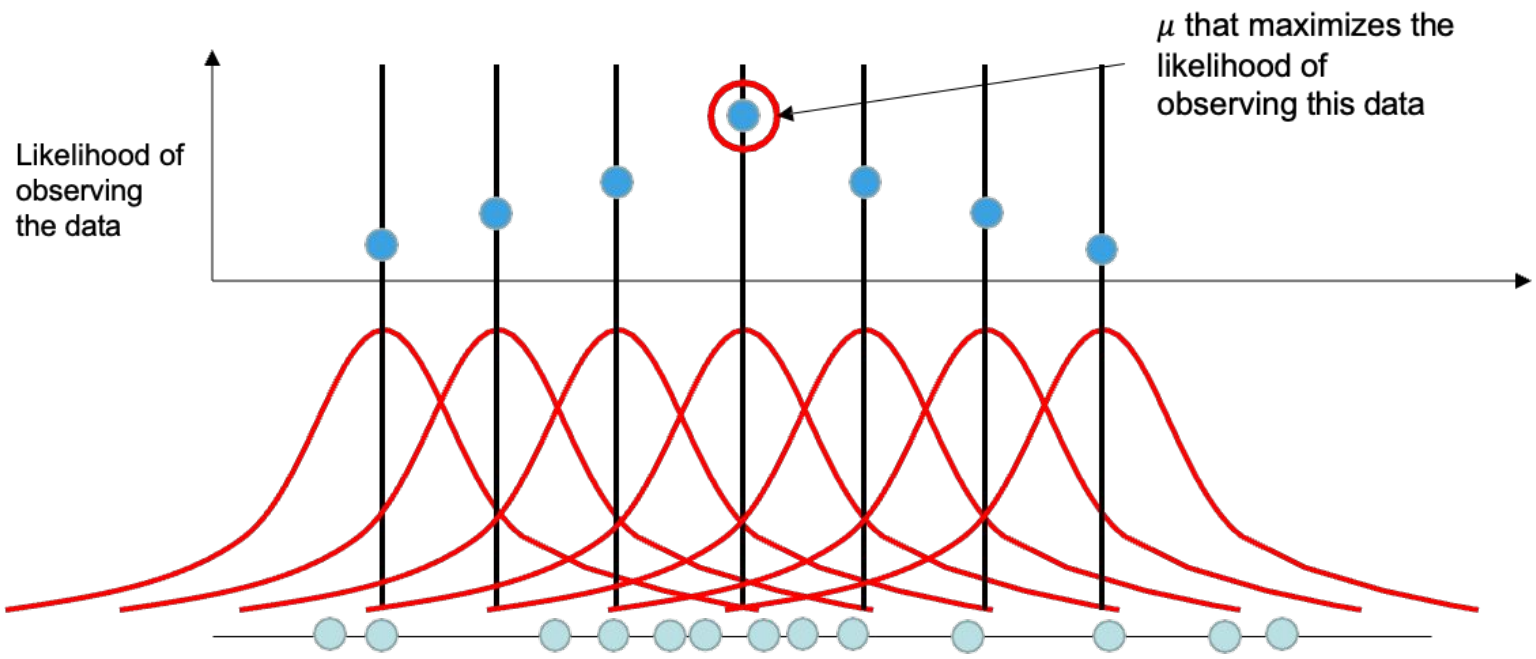
$$L(\boldsymbol{\theta}) \triangleq \log p(\mathcal{D} \mid \boldsymbol{\theta}) = \sum_{n=1}^{N} \log p\left(\mathbf{y}_n \mid \mathbf{x}_n, \boldsymbol{\theta}\right)$$

Then maximize the log likelihood:

$$\hat{\boldsymbol{\theta}}_{\text{mle}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{n=1}^{N} \log p\left(\mathbf{y}_n \mid \mathbf{x}_n, \boldsymbol{\theta}\right)$$

Recall Maximum Likelihood Estimation (MLE)

MLE is a method of estimating the parameters of a statistical model given observations, by finding the parameter values that maximize the likelihood of making the observations



Recall Maximum a Posterior (MAP)

$$\hat{\boldsymbol{\theta}}_{\text{map}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \log p(\boldsymbol{\theta} \mid \mathcal{D}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \log p(\mathcal{D} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$$

$$= \underset{\boldsymbol{\theta}}{\operatorname{argmin}} - \left[\frac{1}{N} \sum_{n=1}^{N} \log p\left(\mathbf{y}_n \mid f\left(\mathbf{x}_n; \boldsymbol{\theta}\right)\right) \right] - \log p(\boldsymbol{\theta})$$

If $p(\boldsymbol{\theta})$ follows a uniform distribution, then MLE = MAP

Solve MLE with Latent Variables

We now describe the setting using mathematical language.

Let $X = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ be our set of d-dimensional observations. We refer to X as our incomplete data. We will assume that the complete data is $D = (X, Z)$

, where $Z = \{\mathbf{z}_1, \ldots, \mathbf{z}_N\}$ represents the N unobserved (hidden) data vectors which are in one-to-one correspondence with X, i.e., $\mathbf{Z}_i$ is associated with $\mathbf{x}_i$.

For example, we can assume Z to be discrete and in the case of mixture models to represent the class labels, i.e., where $\mathbf{z}_i = k$ and $k \in \{1, 2, \ldots, K\}$

Solve MLE with Latent Variables

It maximizes the incomplete-data log-likelihood $\sum_{n=1}^{N} \log p\left(\mathbf{x}_n \mid \boldsymbol{\theta}\right)$

NOT the complete-data log-likelihood $\sum_{n=1}^{N} \log p\left(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}\right)$ as $\mathbf{z}_n$ are hidden variables!

Solve MLE with Latent Variables

Optimization Problem – solve MLE based on the observed data (maximize the incomplete-data log-likelihood)!!!

hard

easy

$$\ell(\boldsymbol{\theta}) = \sum_{n=1}^{N} \log \boxed{p\left(\mathbf{x}_n \mid \boldsymbol{\theta}\right)} = \sum_{n=1}^{N} \log \left[\sum_{\mathbf{z}_n} \boxed{p\left(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}\right)} \right]$$

The incomplete-data log-likelihood can be expressed as the complete-data log-likelihood marginalized over the latent variables.

It is hard to optimize!!

1: The sum is inside the Log. Log of sum calls for nonlinear optimization techniques.

2: The hidden variables $\mathbf{z}_n$, and $\boldsymbol{\theta}$ are unknown.

Part 4: Expectation-Maximization (EM) Algorithm

Expectation-Maximization (EM) Algorithm

Definition: A general iterative optimization method of finding the MLE of the parameters given a dataset with incomplete or missing data (solve MLE with hidden variables).

Where to use: **the data has hidden variables.** For example, 1) the data has missing values or is incomplete; 2) clustering, where the data labels are unknown

Objective: The goal of EM is to maximize the log likelihood of **the observed data**.

High-level Understanding for EM

Understand the EM from several perspectives:

1: Iterative bound optimization

2: Estimate the distributions for hidden variables, then sample hidden data from the distributions to obtain the (approximate) complete data, finally solve MLE based on the complete data

3: Coordinate ascent

It maximizes the incomplete-data log-likelihood $\sum_{n=1}^{N} \log p\left(\mathbf{x}_n \mid \boldsymbol{\theta}\right)$ rather than the complete-data log-likelihood $\sum_{n=1}^{N} \log p\left(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}\right)$ as $\mathbf{z}_n$ are hidden variables!

Bound Optimization

Goal: maximize some objective functions $L(\boldsymbol{\theta})$ that are hard to be solved directly, such as the log likelihood with latent variables wrt its parameters $\boldsymbol{\theta}$.

For the t-th iteration:

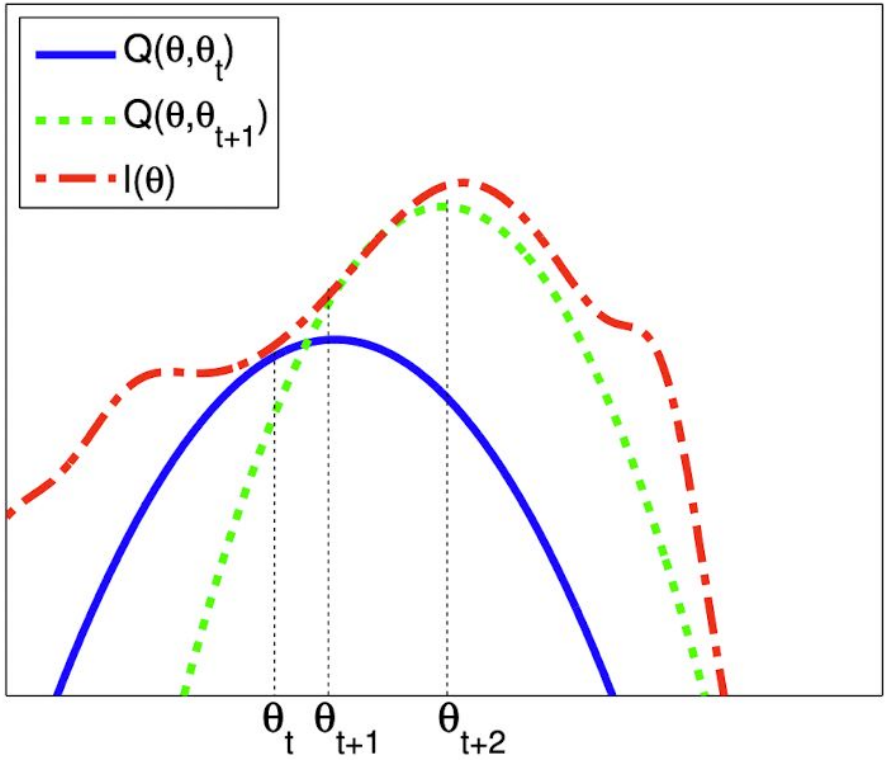
Step 1: construct a surrogate function which is a tight lower bound to $L(\boldsymbol{\theta})$:

$$Q\left(\boldsymbol{\theta}, \boldsymbol{\theta}^t\right) \leq L(\boldsymbol{\theta}) \quad \text{and} \quad Q\left(\boldsymbol{\theta}^t, \boldsymbol{\theta}^t\right) = L\left(\boldsymbol{\theta}^t\right)$$

Step 2: perform the following update:

$$\boldsymbol{\theta}^{t+1} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} Q\left(\boldsymbol{\theta}, \boldsymbol{\theta}^t\right)$$

Bound Optimization



This guarantees the monotonic increases in the original objective!!

$$\ell\left(\boldsymbol{\theta}^{t+1}\right) \geq Q\left(\boldsymbol{\theta}^{t+1}, \boldsymbol{\theta}^{t}\right) \geq Q\left(\boldsymbol{\theta}^{t}, \boldsymbol{\theta}^{t}\right) = \ell\left(\boldsymbol{\theta}^{t}\right)$$

EM Algorithm as Iterative Lower Bound Optimization

Re-written the observed data log likelihood:

$$\ell(\boldsymbol{\theta}) = \sum_{n=1}^{N} \log \left[\sum_{\mathbf{z}_n} q_n(\mathbf{z}_n) \frac{p(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta})}{q_n(\mathbf{z}_n)} \right]$$

Then use Jensen's inequality (page 26) to derive the lower bound as log function is concave:

$$\begin{aligned} \ell(\boldsymbol{\theta}) &\geq \sum_{n} \sum_{\mathbf{z}_n} q_n(\mathbf{z}_n) \log \frac{p(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta})}{q_n(\mathbf{z}_n)} \\ &= \sum_{n} \left[\mathbb{E}_{q_n} \left[\log p(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}) \right] \right] + \mathbb{H}(q_n) \\ &\triangleq Q(\boldsymbol{\theta}, \{q_n\}) \quad \text{Evidence lower bound (ELBO)} \end{aligned}$$

The EM algorithm alternates between maximizing the lower bound Q wrt the distributions $\{q_n\}$ and the parameters $\boldsymbol{\theta}$!!!

Jensen’s inequality

https://en.wikipedia.org/wiki/Jensen%27s_inequality

E-step

To maximize the lower bound $Q(\boldsymbol{\theta}, \{q_n\})$ wrt $\{q_n\}$

$$
\begin{aligned}
Q(\boldsymbol{\theta}, q_n) &= \sum_{\mathbf{z}_n} q_n(\mathbf{z}_n) \log \frac{p(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta})}{q_n(\mathbf{z}_n)} \\
&= \sum_{\mathbf{z}_n} q_n(\mathbf{z}_n) \log \frac{p(\mathbf{z}_n \mid \mathbf{x}_n, \boldsymbol{\theta})\, p(\mathbf{x}_n \mid \boldsymbol{\theta})}{q_n(\mathbf{z}_n)} \\
&= \sum_{\mathbf{z}_n} q_n(\mathbf{z}_n) \log \frac{p(\mathbf{z}_n \mid \mathbf{x}_n, \boldsymbol{\theta})}{q_n(\mathbf{z}_n)} + \sum_{\mathbf{z}_n} q_n(\mathbf{z}_n) \log p(\mathbf{x}_n \mid \boldsymbol{\theta}) \\
&= -\mathbb{KL}(q_n(\mathbf{z}_n) \| p(\mathbf{z}_n \mid \mathbf{x}_n, \boldsymbol{\theta})) + \log p(\mathbf{x}_n \mid \boldsymbol{\theta})
\end{aligned}
$$

To maximize it, we need to set each q_n to the posterior distribution of the hidden variables $\mathbf{Z}_n$ conditioned on the current parameter $\boldsymbol{\theta}$ (the estimate of the last iteration) and the observation $\mathbf{x}_n$

$$
\boxed{q_n^* = p(\boldsymbol{z}_n \mid \boldsymbol{x}_n, \boldsymbol{\theta})}
$$

This makes the lower bound tight!! $Q(\boldsymbol{\theta}, \{q_n^*\}) = \sum_n \log p(\mathbf{x}_n \mid \boldsymbol{\theta}) = \ell(\boldsymbol{\theta})$

KL divergence (relative entropy)

Read Wiki or ask ChatGPT (out of scope)

https://en.wikipedia.org/wiki/Kullback%E2%80%93Leibler_divergence

For discrete probability distributions P and Q defined on the same sample space $\mathcal{X}$, the KL divergence from P to Q is defined to be

$$D_{\mathrm{KL}}(P\|Q) = \sum_{x\in\mathcal{X}} P(x)\log\left(\frac{P(x)}{Q(x)}\right)$$

For continuous distributions P and Q, the KL divergence is

$$D_{\mathrm{KL}}(P\|Q) = \int_{-\infty}^{\infty} p(x)\log\left(\frac{p(x)}{q(x)}\right)dx$$

where p and q denote the probability densities of P and Q.

M-step

In the M step, we need to maximize $Q\left(\boldsymbol{\theta}, \{q_n^*\}\right)$ wrt $\boldsymbol{\theta}$

Given $\{q_n^*\}$, we can now obtain the optimal lower bound function in page 28.

Note that the E step needs to return the full set of posterior distributions

$\{q\left(\mathbf{z}_n\right) : n = 1 : N\}$ (we don't have specific assumptions of the joint probability).

We ignore $\mathbb{H}\left(q_n^*\right)$ as it is a scalar in the M-step ($\{q_n^*\}$ is known)

$$\boxed{Q\left(\boldsymbol{\theta}, \{q_n^*\}\right) = \sum_n \mathbb{E}_{q_n^*(\boldsymbol{z}_n)}\left[\log p\left(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}\right)\right]}$$

M-step

In the M step, we need to maximize $Q\left(\boldsymbol{\theta}, \{q_n^*\}\right)$ wrt $\boldsymbol{\theta}$

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \sum_n \mathbb{E}_{q_n^*(\boldsymbol{z}_n)} \left[\log p\left(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}\right)\right]$$

This is the expected complete data log likelihood. The Q function as the optimal lower bound is the expected complete-data log likelihood. That's why we call it expectation maximization.

EM summarization

- Choose an initial setting for the parameters $\boldsymbol{\theta}^0$
- While the convergence is not met, t ← t + 1:
  - E-step: Obtain $\forall n : q_n^*(\mathbf{z}_n)$ based on $\boldsymbol{\theta}^t$
  - M-step: Obtain $\boldsymbol{\theta}^{\mathrm{t}+1}$ by solving MLE for the expected complete-data log likelihood

Part 5: Mixture Models for Soft Clustering

Mixture Models

High-level idea: Use a form of a weighted sum of simple densities to represent a complex density (assume continuous distribution),

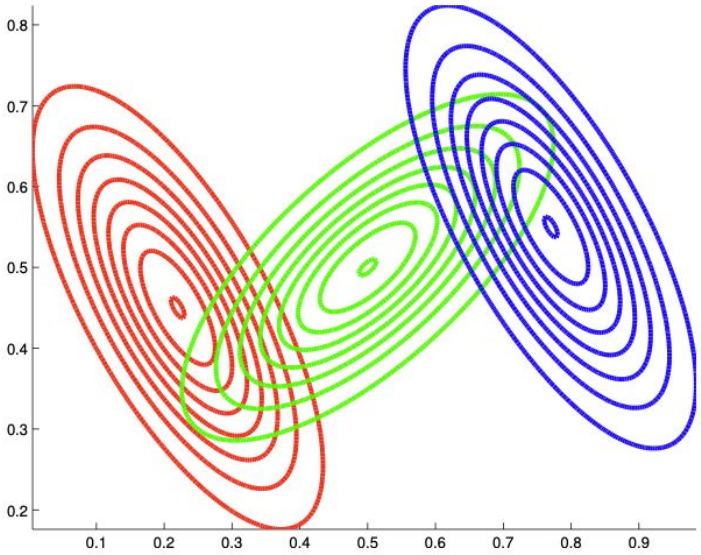
$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^{K} \underbrace{p(\mathbf{x} \mid z = k, \theta)}_{\text{easy}} \underbrace{p(z = k \mid \theta)}_{\text{easy}}$$

$$\sum_{k=1}^{K} p(z = k \mid \theta) = 1 \qquad \int_{\mathbf{x}} p(\mathbf{x} \mid z = k, \theta) d\mathbf{x} = 1$$

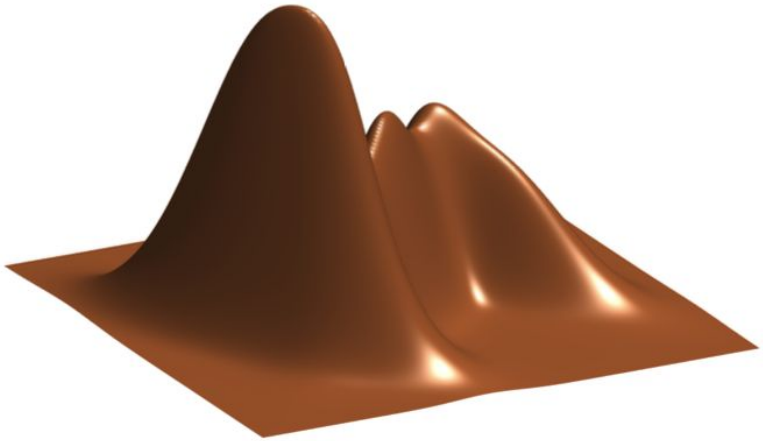
The model decomposes the overall density $p(\mathbf{x} \mid \theta)$ into a weighted linear combination of K components (densities) $p(\mathbf{x} \mid z = k, \theta)$.

The combination coefficient $p(z = k \mid \theta)$ represents the PRIOR of the the k-th component (the probability that a randomly chosen data point is generated by the k-th component).

Mixture Models



(a)



(b)

A mixture of 3 Gaussians in 2d. (a) We show the contours of probability for each component in the mixture. (b) A surface plot of the overall density. Adapted from Figure 2.23 of Bishop's textbook.

Mixture Models

If we know θ and $p(z = k|\theta)$, then we can easily compute the responsibility of a data point **x** in class k given θ (the posterior probability of z=k),

$$p(z = k \mid \mathbf{x}, \theta) = \frac{p(\mathbf{x} \mid z = k, \theta)p(z = k \mid \theta)}{p(\mathbf{x} \mid \theta)} = \frac{p(\mathbf{x} \mid z = k, \theta)p(z = k \mid \theta)}{\sum_{k=1}^{K} p(\mathbf{x} \mid z = k, \theta)p(z = k \mid \theta)}$$

But in practice, both θ and $p(z = k|\theta)$ are unknown!!

Mixture Models

To estimate the parameters, one may want to solve MLE with latent variables that maximizes:

$$\sum_{n=1}^{N} \log p\left(\mathbf{x}_n \mid \theta\right) = \sum_{n=1}^{N} \log \sum_{z_n} p\left(\mathbf{x}_n, z_n \mid \theta\right) = \sum_{n=1}^{N} \log \sum_{k=1}^{K} p\left(\mathbf{x}_n \mid z_n = k, \theta\right) p\left(z_n = k \mid \theta\right)$$

with respect to θ and each $p(z_n = k \mid \theta)$.

Challenges:

1: The unknown parameters enter the maximization problem in a nonlinear fashion

2: The labels of the data points are unknown, that is, the specific class or component from which each data point arises is unknown. If known, we only need to collect the data from each class and then solve K separate MLE estimation.

Estimating the parameters of Mixture Models

Use EM algorithm to estimate the parameters of the components!!

See Week 8